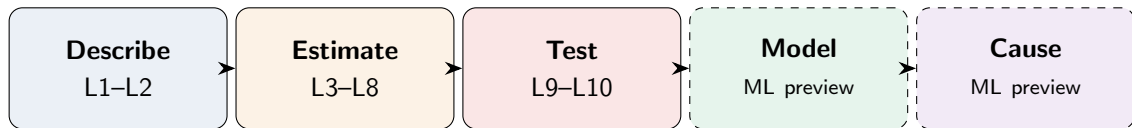


Statistics: The Complete Picture

Everything We Learned in 10 Lectures (+ a Preview of What's Next)

The Journey



L1–L2: What does the data look like? Summaries, plots, pitfalls.

L3–L8: How do we estimate unknown parameters? Properties, MLE, MAP, CIs, bootstrap.

L9–L10: Is the pattern real? Hypothesis tests, ANOVA, A/B testing.

Preview: How do variables relate? Regression, GLMs. (*in ML course*)

Preview: Did *X* cause *Y*? DAGs, potential outcomes, causal inference. (*in ML course*)

L1–L2: Foundations & Descriptive Statistics

L1 — Foundations

- Population vs sample
- Parameter θ vs statistic $\hat{\theta}$
- Estimand / estimator / estimate
- i.i.d. assumption
- Loss functions: L_1 , L_2 , L_0
- Risk = $E[\text{Loss}]$ and ERM
- Plug-in principle

L2 — Descriptive Statistics

- Mean, median, mode
- Variance, IQR, MAD
- Histograms, boxplots, ECDF
- Skewness & kurtosis
- Anscombe's quartet
- Simpson's paradox
- Always plot your data!

Key lesson: Statistics begins with a question about a *population*, answered using a *sample*.

L3–L4: Properties of Estimators

L3 — Bias, Variance, MSE

- Bias = $E[\hat{\theta}] - \theta$
- $\text{MSE} = \text{Bias}^2 + \text{Variance}$
- Bias-variance tradeoff
- Consistency
- Sufficiency (Fisher–Neyman)
- Exponential family

L4 — Fisher Information & CR

- Score function $s(\theta)$
- Fisher info $\mathcal{I}(\theta)$
- Cramér–Rao lower bound:
 $\text{Var}(\hat{\theta}) \geq 1/\mathcal{I}(\theta)$
- Efficiency
- Stein's paradox ($d \geq 3$)

Key lesson: An estimator has a *sampling distribution*. We judge it by bias, variance, and MSE. Fisher information tells us the best we can possibly do.

L5–L6: Point Estimation — MLE & MAP

L5 — MLE

- Method of Moments
- Likelihood $\rightarrow$ log-likelihood
- MLE: $\hat{\theta} = \arg \max \ell(\theta)$
- MLE is consistent, asymp. normal,
efficient (achieves CR bound)
- MLE = least squares (Normal)
- MLE = cross-entropy (logistic)

L6 — MAP & Bayesian

- Prior $\times$ likelihood $\propto$ posterior
- Conjugate priors
- MAP: $\hat{\theta} = \arg \max$ posterior
- **Regularization = MAP:**
 - Gaussian prior $\rightarrow$ Ridge (L_2)
 - Laplace prior $\rightarrow$ Lasso (L_1)
- MLE vs MAP vs full Bayesian

Key lesson: MLE = let the data speak. MAP = data + prior beliefs.
Regularization in ML is Bayesian estimation in disguise.

L7–L8: Sampling Distributions, CIs & Bootstrap

L7 — Sampling Distributions

- $\hat{\theta}$ is a random variable
- Monte Carlo simulation
- CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$
- $SD \neq SE$
- $SE = \sigma/\sqrt{n}$ ($\sqrt{n}$ law)
- SE via Fisher info for MLEs

L8 — CIs & Bootstrap

- Wald CI: $\hat{\theta} \pm z \cdot SE$
- Wilson CI for proportions
- t -interval, delta method
- **Bootstrap:** resample $\rightarrow$ SE, CI
- Percentile & BCa intervals
- Two paths: analytical vs computational

Key lesson: A point estimate without uncertainty is useless.
CIs quantify “how much could the estimate wiggle?”

L9–L10: Hypothesis Testing & Classical Tests

L9 — Hypothesis Testing

- H_0 vs H_1 , test statistic
- Type I / II errors
- p -value: $P(\text{data} \mid H_0)$
- Statistical vs practical significance
- Power & sample size planning
- Permutation tests
- Multiple testing (Bonferroni, BH)

L10 — Classical Tests & LRT

- LRT: $-2 \log \Lambda \rightarrow \chi_k^2$
- One-sample & paired t -test
- Two-sample t (Welch's)
- Test for proportions
- χ^2 goodness-of-fit
- χ^2 test of independence
- Nonparametric: Mann–Whitney, Wilcoxon

Key lesson: A p -value asks “how surprising is the data if H_0 is true?”

Small $p \neq$ large effect. Always report effect sizes.

L10 (cont.): ANOVA & A/B Testing

ANOVA

- Multiple t -tests $\rightarrow$ false positive explosion
- $SS_T = SS_B + SS_W$, $F = MS_B/MS_W$
- Post-hoc: Tukey HSD, Bonferroni
- Effect size η^2 ; two-way ANOVA

A/B Testing

- A/B test = randomized experiment
- MDE, sample size planning
- Pitfalls: peeking ($>25\%$ false pos.),
too many metrics, novelty
- Sequential testing (O'Brien-Fleming)

Key lesson: Comparing k groups needs ANOVA, not $\binom{k}{2}$ t -tests. A/B testing = hypothesis testing for products.

Preview: Regression Inference & GLMs (*in ML course*)

L12 — Regression Inference

- $Y = X\beta + \varepsilon$, LINE assumptions
- SE of $\hat{\beta}_j$, t -tests, CIs
- Overall F -test, R^2 , R^2_{adj}
- Gauss–Markov: OLS is BLUE
- Diagnostics: residuals, QQ, leverage, Cook's distance
- Logistic: Z -tests, odds ratios

L13 — GLMs

- Three components:
Random + Systematic + Link
- Exponential family
- OLS = GLM(Normal, Identity)
- Logistic = GLM(Bernoulli, Logit)
- Poisson regression (log link)
- IRLS, deviance, AIC

Key lesson: OLS, logistic, and Poisson regression are all GLMs — same framework, different distribution and link function.

Preview: Causal Inference (*in ML course*)

The Problem

- Correlation $\neq$ causation
- Confounding, reverse causation, collider bias
- Potential outcomes: $Y(1), Y(0)$
- Fundamental problem: only one is observed
- $ATE = E[Y(1) - Y(0)]$

The Solutions

- RCTs / A/B tests (gold standard)
- DAGs: fork, chain, collider
- Backdoor criterion
- $do(X) \neq \text{observe } X$
- Observational methods: regression, matching, propensity scores, IV

Key lesson: To claim causation, you need either randomization or causal assumptions (encoded in a DAG). Data alone is not enough.

The Formulas That Matter Most

MSE decomposition

$$\text{MSE} = \text{Bias}^2 + \text{Var}$$

MLE

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \sum_i \log f(x_i | \theta)$$

Confidence interval

$$\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}(\hat{\theta})$$

GLM

$$g(\mu) = X\beta, \quad Y \sim \text{exp. family}$$

Cramér–Rao bound

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n \mathcal{I}(\theta)}$$

MAP

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\ell(\theta) + \log \pi(\theta)]$$

Likelihood ratio test

$$-2 \log \frac{L(\hat{\theta}_0)}{L(\hat{\theta})} \sim \chi_k^2$$

ATE (causal)

$$\text{ATE} = E[Y(1)] - E[Y(0)]$$

Five Themes That Connect Everything

1. Uncertainty is the point. Every estimate has a sampling distribution. Report SE, CI, p -value — not just the number.

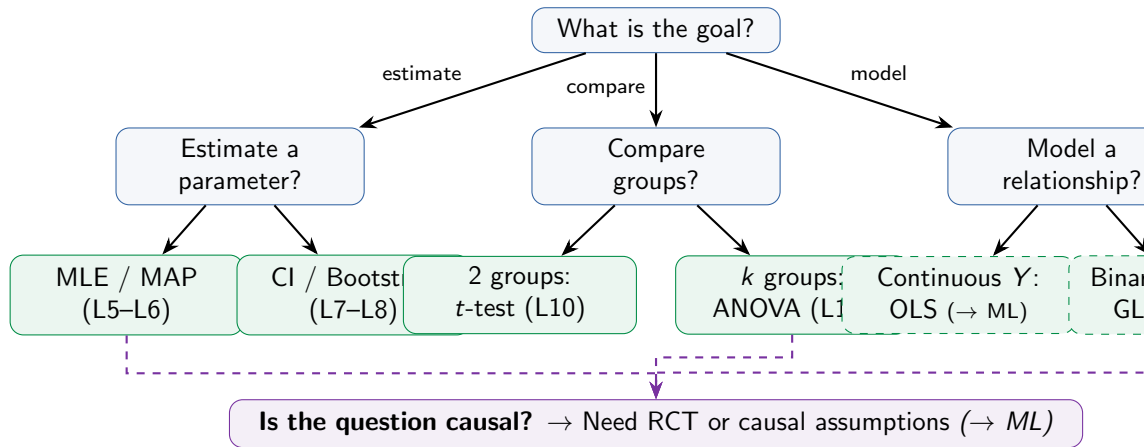
2. Bias-variance tradeoff is everywhere. MLE vs MAP. OLS vs ridge. Simple vs complex model. More data helps variance, not bias.

3. Likelihood is the universal language. MLE, LRT, Fisher info, deviance, AIC, GLMs — all built on $L(\theta \mid \text{data})$.

4. Assumptions matter more than methods. Every test and model has assumptions (normality, independence, linearity, no confounding). Violations matter.

5. Association $\neq$ causation. Regression finds patterns. Only experiments (or careful causal reasoning) find causes.

Which Method Do I Use?



Top Mistakes to Avoid

✗ Confusing SD and SE

SD = spread of data.

SE = spread of $\hat{\theta}$. $SE \rightarrow 0$ as $n \rightarrow \infty$.

✗ Ignoring assumptions

t -test on heavy-tailed data, OLS on binary Y , ANOVA with unequal variances.

✗ Correlation $\rightarrow$ causation

Regression finds association.

Causation needs a DAG or an experiment.

✗ “ $p > 0.05 = \text{no effect}$ ”

Absence of evidence $\neq$ evidence of absence. Could be low power.

✗ Peeking at p -values

Testing repeatedly as data arrives inflates false positives to $>25\%$.

✗ R^2 worship

R^2 always increases with more predictors.

Use R^2_{adj} or AIC. Check diagnostics.

Your Python Toolkit

Task	Library	Key function
Descriptive stats	numpy, pandas	.describe(), np.quantile()
Plotting	matplotlib, seaborn	sns.boxplot(), sns.histplot()
MLE	scipy.stats	.fit(), minimize(-loglik)
Bootstrap	scipy.stats	bootstrap()
t -tests	scipy.stats	ttest_ind(), ttest_rel()
χ^2 tests	scipy.stats	chi2_contingency()
ANOVA	scipy.stats	f_oneway()
OLS regression (<i>ML</i>)	statsmodels	OLS().fit().summary()
GLMs (<i>ML</i>)	statsmodels	GLM(family=...).fit()
Logistic (<i>ML</i>)	statsmodels	Logit().fit()
Causal inference (<i>ML</i>)	dowhy	CausalModel()
Prop. scores (<i>ML</i>)	causalml	PropensityScoreMatching()

Where to Go Next

Machine Learning

Bias-variance,
cross-validation,
regularization, trees,
neural networks

Bayesian Statistics

MCMC, Gibbs,
hierarchical models,
Bayesian regression,
PyMC, Stan

Time Series

AR, MA, ARIMA,
stationarity,
forecasting,
statsmodels.tsa

Causal ML

Heterogeneous
treatment effects,
causal forests,
double ML

Experimental Design

Block designs,
factorial experiments,
multi-armed bandits,
adaptive trials

That's Statistics!

14 lectures · From data to causation

"All models are wrong, but some are useful."

— George E. P. Box